(a) $(0.5 \cdot \text{CLIP}(\text{“}\textit{cat}\text{”})) + (0.5 \cdot \text{CLIP}(\text{“}\textit{tree}\text{”}))$

(b) $(0.5 \cdot \text{CLIP}(\text{“}\textit{dog}\text{”})) + (0.5 \cdot \text{CLIP}(\text{“}\textit{lake}\text{”}))$

(c) $(0.5 \cdot \text{CLIP}(\text{“}\textit{bear}\text{”})) + (0.5 \cdot \text{CLIP}(\text{“}\textit{hat}\text{”}))$

(d) $(0.5 \cdot \text{CLIP}(\text{“}\textit{elephant}\text{”})) + (0.5 \cdot \text{CLIP}(\text{“}\textit{snow}\text{”}))$

(e) $(0.5 \cdot \text{CLIP}(\text{“}\textit{goat}\text{”})) + (0.5 \cdot \text{CLIP}(\text{“}\textit{crown}\text{”}))$

(f) $(0.5 \cdot \text{CLIP}(\text{“}\textit{tiger}\text{”})) + (0.5 \cdot \text{CLIP}(\text{“}\textit{glasses}\text{”}))$

Figure 3: Images generated by summing representations of multiple prompts

of the contents of the images generated for each pair of prompts, along with details of statistical significance tests are given in App. B.1.

There is no particular reason to assume that summing the representations of two words would result in a representation that can be interpreted by the model. Adding the representations in this way could result in a representation that doesn't correspond to anything at all, and leads to the generation of low quality images. It's also plausible that it would produce an interpolation between the two concepts described. However, we can see that this is not the case. The model sometimes produced both of the objects described in the two prompts, and otherwise largely produced one or the other. This indicates that the representations are encoded in such a way that the important information corresponding to each prompt does not interfere when the two are summed, potentially because they are stored in directions that are close to orthogonal.

As an additional observation on this point,

Input	Percentage containing: $\mathbf{s}_1$	$\mathbf{s}_2$	Both	Neither
$\text{CLIP}(\mathbf{s}_1)$	100.0	0	0	0
$\text{CLIP}(\mathbf{s}_2)$	0	100.0	0	0
$0.5 \cdot (\text{CLIP}(\mathbf{s}_1)) + 0.5 \cdot (\text{CLIP}(\mathbf{s}_2))$	35.2	20	41.5	3.3

Table 1: Table showing which prompts were represented in images generated from each prompt in a pair and from their weighted sum

we found that some pairs, such as $\mathbf{s}_1$ =“*cat*”, $\mathbf{s}_2$ =“*dog*”, did not behave in the same way, instead producing a cat-dog hybrid creature. We speculate that this may be because similar concepts such as “*cat*” and “*dog*” are likely to share similar representation space.

4 Claim 2: Polysemous words are represented as linear superpositions of meanings

Next, we seek to show that the CLIP encoder used by Stable Diffusion represents polysemous word as a linear superposition of their possible meanings. Other methods of encoding words have been previously shown to represent polysemy in this way (Arora et al., 2018). In this case, since the encoder is transformer-based, there is no static representation of each word, but instead a representation of each token of the word in the context of a sentence. Due to this, we instead approximate directions corresponding to each meaning across several sentences. For a particular word of interest, we take several sets of syntactically similar sentences where each sentence contains either an ambiguous use of the word, or a use where one sense is more likely than the other. A procedure similar to that used by Bolukbasi et al. (2016) is then used to obtain bases for the spaces corresponding to the difference between a representation of a particular meaning and the representation of the word in the corresponding ambiguous sentence. We then use projections into these spaces to obtain approximations of the representation of each meaning. We then demonstrate how these approximations can be used to intervene on a representation and change the sense that appears in generated images. This involves editing the representation in the subspace spanned by our approximate representations, and removing the contribution from the representation of the undesired meaning.

(a) Unedited prompt encoding

(b) Encoding edited to favour the animal sense

(c) Encoding edited to favour the sports-related sense

Figure 4: Prompt: "*a bat laying on the grass*"

(a) Unedited prompt encoding

(b) Encoding edited to favour the music-related sense

(c) Encoding edited to favour the fish sense

Figure 5: Prompt: "*a man holding a bass*"

4.1 Identifying a Meaning Subspace

In a procedure based on that described by Bolukbasi et al. (2016), we begin by identifying a subspace in which representations of different meanings differ. Given a polysemous word w_p with distinct possible meanings, m_1 and m_2, we hand-write a set of N prompts containing w_p in which either of the two candidate meanings is possible. From each such prompt, $\mathbf{s}^{\text{amb}}$, we manually create $\mathbf{s}^{(1)}$ and $\mathbf{s}^{(2)}$, which differ from $\mathbf{s}^{\text{amb}}$ by the addition of a disambiguating modifier and changes to the lexical content to more strongly imply meanings m_1 and m_2 respectively, but otherwise have the same syntactic structure. For example,

$$\mathbf{s}^{\text{amb}} = \text{“}\textit{a bat in a box}\text{”}$$
$$\mathbf{s}^{(1)} = \text{“}\textit{a vampire bat in a cave}\text{”}$$
$$\mathbf{s}^{(2)} = \text{“}\textit{a baseball bat in a locker}\text{”}$$

Each sentence is encoded using CLIP and the vector representing w_p is extracted from each to give $V^{\mathbf{s}_n}_{w_p} = \left\{\mathbf{v}^{\mathbf{s}^{\text{amb}}_n}_{w_p}, \mathbf{v}^{\mathbf{s}^{(1)}_n}_{w_p}, \mathbf{v}^{\mathbf{s}^{(2)}_n}_{w_p}\right\}$, where $\mathbf{v}^{\mathbf{s}}_{w_p} = \text{CLIP}(\mathbf{s})_{w_p}$.[3]

Next we construct $D^{\text{amb}\to 1}_n = \left\{\mathbf{v}^{\mathbf{s}^{\text{amb}}_n}_{w_p}, \mathbf{v}^{\mathbf{s}^{(1)}_n}_{w_p}\right\}$, and calculate the mean of each such set:

$$\boldsymbol{\mu}^{\text{amb}\to 1}_n = \frac{1}{|D^{\text{amb}\to 1}_n|} \sum_{\mathbf{v}\in D^{\text{amb}\to 1}_n} \mathbf{v} \tag{1}$$

A matrix is calculated by summing the outer products of each vector's difference with the mean of its set:

$$C^{\text{amb}\to 1} = \sum_{n=1}^{N} \sum_{\mathbf{v}\in D^{\text{amb}\to 1}_n} (\mathbf{v} - \boldsymbol{\mu}^{\text{amb}\to 1}_n)(\mathbf{v} - \boldsymbol{\mu}^{\text{amb}\to 1}_n)^\top \tag{2}$$

The singular value decomposition (SVD) gives $C^{\text{amb}\to 1} = U_1 \Sigma_1 V_1^\top$. The first k columns of the matrix U_1, $\{\mathbf{u}^{\text{amb}\to 1}_1, \ldots, \mathbf{u}^{\text{amb}\to 1}_k\}$ give an orthonormal basis for the subspace of difference between $\mathbf{v}^{\mathbf{s}^{\text{amb}}}_{w_p}$ and $\mathbf{v}^{\mathbf{s}^{(1)}}_{w_p}$.[4] To find $\{\mathbf{u}^{\text{amb}\to 2}_1, \ldots, \mathbf{u}^{\text{amb}\to 2}_k\}$, this procedure is repeated starting with $D^{\text{amb}\to 2}_n = \left\{\mathbf{v}^{\mathbf{s}^{\text{amb}}_n}_{w_p}, \mathbf{v}^{\mathbf{s}^{(2)}_n}_{w_p}\right\}$.

4.2 Finding Meaning Directions

We then use the difference subspaces defined by these bases to approximate vectors corresponding to m_1 and m_2, which we term $\mathbf{v}_1$ and $\mathbf{v}_2$.

[3] In practice, all of the words we used in our experiments were tokenised to a single token, so our description assumes that we have one encoding per word. For a word consisting of multiple tokens, this process would be repeated for the encoding corresponding to each token.

[4] We choose k based on a threshold on the corresponding singular values, specifically we take k to be the smallest integer greater than 2 such that $\frac{\sum_{i=1}^{k}\sigma_i}{\sum_{i=1}^{N}\sigma_i} > 0.95$, where σ_i is the ith diagonal entry of Σ_1. In practice, for all of our examples this results in $k < 5$.

The procedure for approximating these vectors begins by finding the average projection of $\mathbf{v}_{w_p}^{\mathbf{s}^{(1)}}$ into the subspace defined by $\{\mathbf{u}_1^{\text{amb}\to 1}, \ldots, \mathbf{u}_k^{\text{amb}\to 1}\}$:

$$\mathbf{v}_1^{(0)} = \sum_{j=1}^{k} \left(\frac{1}{N} \sum_{n=1}^{N} \mathbf{v}_{w_p}^{\mathbf{s}_n^{(1)}} \cdot \mathbf{u}_j^{\text{amb}\to 1} \right) \mathbf{u}_j^{\text{amb}\to 1} \tag{3}$$

We then make this orthogonal to all $\mathbf{v}_{w_p}^{\mathbf{s}^{(2)}}$, by taking $\widehat{\mathbf{v}_1}$ to be $\frac{\mathbf{v}_1^{(N)}}{||\mathbf{v}_1^{(N)}||}$, where

$$\mathbf{v}_1^{(n)} = \mathbf{v}_1^{(n-1)} - \text{proj}(\mathbf{v}_1^{(n-1)}, \mathbf{v}_{w_p}^{\mathbf{s}_n^{(2)}}) \tag{4}$$

where $\text{proj}(\mathbf{a}, \mathbf{b})$ denotes the projection of $\mathbf{a}$ onto $\mathbf{b}$, i.e. $\text{proj}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{b}\cdot\mathbf{b}}\mathbf{b}$. The vector $\widehat{\mathbf{v}_2}$ is found analogously.

By examination, we find that representations from ambiguous sentences tend to have negative dot products with both $\widehat{\mathbf{v}_1}$ and $\widehat{\mathbf{v}_2}$ (or occasionally positive dot products with both). Representations from sentences favouring m_1 tend to have a a positive dot product with $\widehat{\mathbf{v}_1}$ and a near-zero dot product with $\widehat{\mathbf{v}_2}$ (by construction), and vice versa for sentences favouring m_2. This suggests how we may edit representations to target a desired sense.

The normalised vectors $\widehat{\mathbf{v}_1}$ and $\widehat{\mathbf{v}_2}$ are not used, instead we take $\mathbf{v}_1 = \max_n(\mathbf{v}_{w_p}^{\mathbf{s}_n^{(1)}} \cdot \widehat{\mathbf{v}_1})\widehat{\mathbf{v}_1}$ and $\mathbf{v}_2 = \max_n(\mathbf{v}_{w_p}^{\mathbf{s}_n^{(2)}} \cdot \widehat{\mathbf{v}_2})\widehat{\mathbf{v}_2}$. We do this so that the vector also represents the magnitude of projection is this direction that is sufficient to have this sense represented.

4.3 Sense Editing Procedure

If we have a sentence $\mathbf{s}$ containing w_p then its standard, unedited encoding is given by $\mathbf{v}_{w_p} = \text{CLIP}(\mathbf{s})_{w_p}$. To edit this representation to push its sense towards m_2, we consider its projection onto the subspace spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$ and aim to make its dot product with $\mathbf{v}_1$ in this subspace near-zero while aligning it with $\mathbf{v}_2$, based on the observations described in the previous section.

Since $\mathbf{v}_1$ and $\mathbf{v}_2$ are, in general, not orthogonal, we first obtain an orthonormal basis of the subspace that they span: $U = \left\{ \frac{\mathbf{v}_1}{||\mathbf{v}_1||}, \frac{\mathbf{v}_2 - \text{proj}(\mathbf{v}_2, \frac{\mathbf{v}_1}{||\mathbf{v}_1||})}{||\mathbf{v}_2 - \text{proj}(\mathbf{v}_2, \frac{\mathbf{v}_1}{||\mathbf{v}_1||})||} \right\}$, and then compute

$$\widetilde{\mathbf{v}}_{w_p}^{m_2} = \mathbf{v}_{w_p} - \text{proj}_U \mathbf{v}_{w_p} + \frac{||\mathbf{v}_2||}{||\mathbf{v}_2 - \text{proj}(\mathbf{v}_2, \mathbf{v}_1)||}(\mathbf{v}_2 - \text{proj}(\mathbf{v}_2, \mathbf{v}_1)) \tag{5}$$

Prompts	
a **bass**	a **crane**
a man holds a **bass**	a **crane** by the ocean
a **bass** is displayed on a wall	a **crane** surrounded by nature
a **bat**	**glasses** on a table
a **bat** and a baseball fly through the air	a **seal**
a boy holds a black **bat**	a **seal** on an envelope
a **bat** laying on the grass	a **trunk**

Table 2: Prompts used to evaluate our editing procedure, with the word of interest in bold

where $\text{proj}_U \mathbf{v}$ is the projection onto the subspace defined by U, i.e. $\text{proj}_U \mathbf{v} = \sum_{\mathbf{u}\in U}(\mathbf{v}\cdot\mathbf{u})\mathbf{u}$.

Subtracting $\text{proj}_U \mathbf{v}_{w_p}$ removes the entire projection of the vector onto this subspace, including its component along $\mathbf{v}_1$, making it orthogonal to $\mathbf{v}_1$ as desired. We then wish to add $\mathbf{v}_2$, but since this is not orthogonal to $\mathbf{v}_1$ we remove its projection along $\mathbf{v}_1$ and rescale accordingly. This means that in the subspace defined by U, $\widetilde{\mathbf{v}}_{w_p}^{m_2}$ is orthogonal to $\mathbf{v}_1$ and has high cosine similarity with $\mathbf{v}_2$. This edited representation replaces $\mathbf{v}_{w_p}$ in the sentence encoding, which is then passed to the model as input.

4.4 Experimental Results

The sense editing procedure was performed on a set of prompts (shown in table 2) that were selected either because they had been observed to exhibit homonym duplication or because the sense that was represented in generated images was not always the same. In Fig. 4 and Fig. 5, we show some example images generated using an unedited prompt and some using prompts edited to favour one sense over another. More images are provided in App. A.3. In Fig. 6, we display the proportion of images realising each sense before and after applying this procedure (with $\rightarrow$ Sense 1 and $\rightarrow$ Sense 2 denoting the encodings edited to favour Sense 1 and Sense 2 respectively). Detailed numbers and details of statistical significance testing can be found in App. B.2. We can see that this procedure is successful in shifting the represented sense towards the intended sense. This suggests that the representations produced by CLIP do indeed incorporate a linear sum over representations of possible meanings, because a representation can be rewritten in terms of its projection into the subspace spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$ and a component in the nullspace of this projection. It also shows that manipulating this sum affects the content of the resulting images.